

A Label-Efficient Self-Supervised Learning Method for Sleep Stage Classification in Multi-channel In-home EEG Signals

Abstract—Automatic sleep stage classification is an important task to assist experts to perform diagnosis of sleep-related disorders. In supervised learning setting, the availability of large number of labeled training data is often the bottleneck for training the sleep stage classification model. Furthermore, the advent of multi-channel EEG signal from in-home EEG devices for sleep monitoring also pose a challenge of designing a label-efficient model suitable for such data. This work proposes a label-efficient self-supervised learning approach for performing 5 sleep stage classification suitable for multi-channel in-home EEG device signal. Our work demonstrates the effectiveness of employing contrastive learning technique on unlabeled EEG data to learn the prominent features. We applied fine-tuning of contrastively learned features to perform sleep stage classification using limited number of labeled data. The experimental results suggest that the proposed approach is more effective than training a traditional end-to-end sleep stage classification model without contrastively learned features. Specially, our approach is suitable in the situation where large number of unlabeled data is available and a small number of labeled data is provided. Furthermore, our results suggest that the proposed method predicts with higher confidence than end-to-end model.

Index Terms—In-home EEG device, Multi-channel EEG signal, Sleep stage classification, Contrastive learning, Self-supervised learning

I. INTRODUCTION

Sleep is an essential part of human life and its disruption may cause numerous health risks such as cardiovascular disease [10], obesity [12], depression [14], and memory loss [6]. Due to the immense importance of sleep for human well-being, recording and analyzing biological signals are often carried out to diagnose sleep disorder. In clinical settings, Polysomnography (PSG) is employed to record the three types of biological signals: electroencephalogram (EEG), electromyogram (EMG) and electrooculogram (EOG) recorded using 6, 4 and 2 electrodes or channels, respectively. The main task of the sleep experts is to identify 30-seconds segments of the PSG signal as one of the 5 sleep stages - wake (W), rapid eye movement (REM) and non-REM 1-3 (NREM1-3) based on the occurrence of characteristic waves [1]. Subsequently, the transition from one stage to another can be analyzed to diagnose sleep disorders such as insomnia, narcolepsy and parasomnia —. However, one of the main drawbacks of PSG is that it can be carried out in clinics or hospitals by experts resulting in limited scalability of sleep recordings.

Recently, portable in-home EEG recording devices have been developed that can record sleep using lesser number of channels than PSG. Therefore, in-home EEG devices are

user-friendly and scalable alternative to PSG. Nevertheless, the overall process of analyzing the signals to perform 5 sleep stage scoring is a challenging task, and automatic scoring is often required.

To address the problem of automatic sleep stage scoring, numerous deep neural network (DNN) based methods have been proposed considering the supervised learning settings. In fact, DNNs require large number of data for training. But, obtaining labeled sleep data is a laborious and time-consuming task which can only be performed by sleep experts.

Generally, supervised learning methods are label inefficient due to their high dependence on the quality and quantity of training data [8]. In contrast, self-supervised learning techniques are label-efficient because of their ability to learn important patterns from unlabeled data [5] [7]. Therefore, they are more appealing in the situations where a large number of unlabeled data and small number of labeled data is available. Similarly, in the context of sleep stage scoring, it is easier to obtain small number of labeled data from large unlabeled sleep data. Hence, self-supervised learning can be useful for training the DNN to learn general patterns from unlabeled data and then the learned features can be fine-tuned to perform sleep stage classification.

Contrastive learning is one of the self-supervised learning techniques. Basically, it learns from unlabeled data by considering two different augmentations of a sample as positive and augmentations of other samples as negative augmentations. The model learns using the contrastive loss function in order to increase the similarity between positive pairs and decrease the similarity between positive and negative pairs [4]. Thus, the patterns or features of data are learned in a self-supervised fashion.

There are some existing works which use contrastive learning for sleep stage score [8] [11]. However, most of these works are using single channel from PSG data. Different from the existing works, we are employing contrastive learning on a 3-channel input signal from in-home EEG device.

This research targets the problem of label-efficient sleep stage classification for 3-channel in-home EEG device. More specifically, this research employs contrastive learning technique to extract features from unlabeled data in feature extraction phase. After that, the contrastively learned features are fine-tuned for sleep stage classification task using limited number of labeled data. In the experiments, we compared fine-tuning the model (with contrastively learned features) and

training the model from scratch (end-to-end learning model). Our results suggest that contrastive learning is advantageous for learning features from unlabeled data. For sleep stage classification, We observed that fine-tuning the features learned using contrastive learning is better than feature learning from scratch especially when the labeled training data are limited.

The main contributions of this work are summarized as follows:

- 1) We demonstrate the application of contrastive learning for 3-channel in-home EEG device data.
- 2) We perform the experimental evaluation to show the benefit of fine-tuning the feature extraction layers trained using contrastive learning over end-to-end supervised learning.

II. RELATED WORK

Automatic sleep stage classification is a challenging task. Deep neural networks (DNN) offer a promising solution to this problem due to their ability to learn complex data patterns [17].

Most of the existing works employ DNN for sleep stage classification in supervised learning settings to improve the prediction accuracy on PSG data. Methods like SleepNet [2], Utime [15], and others [3] [18] [19] train convolutional neural networks [9] using labeled PSG data. SleepEEGNet [13] solved the problem of using single channel EEG signal. — further improved the accuracy and introduced the feature of explainability of the predicted sleep score using class activation mapping in 3-channel PSG data. However, the performance of the existing methods strictly depend on the quality and the quantity of labeled training data.

Self-supervised learning methods, particularly contrastive learning do not require large amount of labeled training data. In general, contrastive learning can learn prominent data patterns in self-supervised fashion using unlabeled data, and requires a small number of labeled data to perform the downstream task e.g. classification of images [4].

Recently, contrastive learning has been applied to sleep data. In [11], single channel EEG signal obtained from PSG was used to train a large neural network using contrastive learning. After contrastive training, the feature extraction layers are fine-tuned to perform sleep stage classification using a small number of labeled data. Eldele et. al. in [8] showed a comprehensive evaluation of various self-supervised learning techniques (including contrastive learning) and different kinds of deep neural network architectures for sleep stage classification. However, their evaluation was focused on single channel EEG signal from PSG data.

The existing works employing contrastive learning do not consider the contrastive training of the model using multichannel. Furthermore, the existing methods consider PSG data, whereas it is important to explore the application of contrastive learning for in-home EEG device data as they are gradually becoming common in monitoring sleep and diagnosis of sleep-related disorders. This research attempts to propose an effective method for leveraging contrastive learning

for self-supervised feature extraction from multichannel in-home EEG sleep data. This work shows that features learned using contrastive learning from in-home multichannel EEG device data are effective for enabling label-efficient model fine-tuning for performing sleep stage classification.

III. METHOD

A. Contrastive Learning

Contrastive learning is a type of self-supervised learning that learns the pattern existing in the training examples by comparing the similarity and dissimilarity of the examples from the others. In essence, the model's task is to learn to project similar examples closer to each other than dissimilar examples. In order to learn to discriminate similar and dissimilar examples, the model learns the prominent patterns of the data without requiring labels. Hence, the general features of the data can be learned in a self-supervised fashion. Next, the learned features can be fine-tune for the downstream task in supervised learning settings using a limited number of labeled data. This makes contrastive learning favorable in the situation where we are given a large amount of unlabeled data, and a limited number of labeled data is available.

In this study, we employ the contrastive learning framework to extract robust features from multichannel EEG signals from the in-home EEG device for monitor human sleep. More concretely, in sleep studies, an epoch refers to a 30-second segment of the EEG signal. Given an epoch, two different data augmentation techniques are applied resulting in a positive pair, whereas augmentation of the other epochs are considered negative pairs. The model learns to maximize the similarity between the same positive pairs while minimizing their similarity to the negative pairs. The loss is calculated with a contrastive loss function that accounts for the similarity between positive and negative pairs.

As illustrated in Figure 3, our contrastive learning framework consists of the following core components. Also, the overall contrastive learning algorithm is given in Algorithm 1.

Data Augmentation: Given a minibatch B of N epochs and a set of predefined transformations T , each epoch $x_k \in B$ is transformed into two augmented views, $\tilde{x}_{2k-1}$ and $\tilde{x}_{2k}$. These transformations generate positive pairs of the given epoch x_k while preserving the critical characteristics of the EEG signals. The details of the augmentations used in this study are given in Section III-B. Hence, for B of size N , $2N$ augmentations are generated.

Feature Extractor: Both augmented views $\tilde{x}_{2k-1}$ and $\tilde{x}_{2k}$ pass through a shared feature extractor, denoted as $f(\cdot)$. The feature extractor captures the spatial features of the EEG signals and maps them to an intermediate feature space.

Projection Head: The project head $g(\cdot)$ is responsible for projecting the output features h_{2k-1} and h_{2k} of the positive pair close to each other, and the negative pairs should be projected away from the positive pair. The projection head maps the features into a latent space where the contrastive loss is applied, ensuring that the representations are optimized for

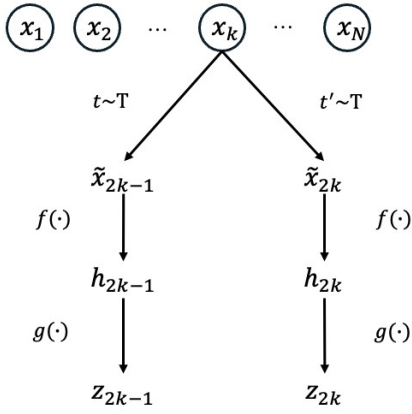


Fig. 1: Overall structure of the contrastive learning framework. The input $\mathbf{x}$ represents a sequence of EEG signals containing N epochs. x_k represents one epoch input EEG signal, where $k \in \{1, 2, \dots, N\}$. $\tilde{x}_{2k-1}$ and $\tilde{x}_{2k}$ are the augmented views generated by the data augmentation pipeline. h_{2k-1} and h_{2k} denote the intermediate representations obtained from the feature extractor $f(\cdot)$, and z_{2k-1} , z_{2k} are the final representations produced by the projection head $g(\cdot)$.

the alignment of positive pairs and the separation of negative pairs.

Contrastive Loss Function: The representations z_{2k-1} and z_{2k} obtained from the projection head are used to calculate the contrastive loss function. Given $x_k \in B$, the contrastive loss w.r.t. x_k can be calculated as given in Equation 1.

$$\ell_{2k-1,2k} = -\log \frac{\exp(\text{sim}(z_{2k-1}, z_{2k})/\tau)}{\sum_{j=1}^{2N} \mathbb{I}_{[j \neq 2k-1]} \exp(\text{sim}(z_{2k-1}, z_j)/\tau)}, \quad (1)$$

where τ is a temperature parameter that controls the sharpness of the similarity distribution, and $\mathbb{I}_{[j \neq 2k-1]}$ is an indicator function. Hence, the total contrastive loss for a batch is computed:

$$\mathcal{L} = \frac{1}{2N} \sum_{k=1}^N (\ell_{2k-1,2k} + \ell_{2k,2k-1}). \quad (2)$$

The similarity between two representations is measured using cosine similarity, defined as:

$$\text{sim}(z_i, z_j) = \frac{z_i^\top z_j}{\|z_i\| \|z_j\|}. \quad (3)$$

The loss function encourages the model to project positive pairs z_{2k-1} and z_{2k} closer, and push apart pairs of z_{2k-1} and z_{2k} with other views in B . Consequently, $2N - 2$ negative pairs are considered for each positive pair to calculate the contrastive loss. The overall contrastive loss for the minibatch (Eq. 2) is used for training the model.

Algorithm 1 Contrastive Learning for EEG Signal

Require: Batch size N , temperature τ , feature extractor $f(\cdot)$, projection head $g(\cdot)$, augmentation operators $t, t' \sim T$

- 1: Initialize parameters of $f(\cdot)$ and $g(\cdot)$
- 2: **for** each training iteration **do**
- 3: Sample a minibatch of N EEG signal epochs $\{x_k\}_{k=1}^N$
- 4: **for** each x_k in minibatch **do**
- 5: data augmentation:

$$\tilde{x}_{2k-1} = t(x_k), \quad \tilde{x}_{2k} = t'(x_k)$$
- 6: feature extractor:

$$h_{2k-1} = f(\tilde{x}_{2k-1}), \quad h_{2k} = f(\tilde{x}_{2k})$$
- 7: projection head:

$$z_{2k-1} = g(h_{2k-1}), \quad z_{2k} = g(h_{2k})$$
- 8: **end for**
- 9: Compute pairwise similarities using Eq. 4.
- 10: Compute the contrastive loss $\mathcal{L}$ for the batch using Eq. 2.
- 11: Update parameters of $f(\cdot)$ and $g(\cdot)$ to minimize $\mathcal{L}$
- 12: **end for**
- 13: **Output:** Trained feature extractor $f(\cdot)$

B. Data Augmentation Methods

Data augmentation is a critical component of the contrastive learning pipeline, as it generates the variability needed to create positive and negative pairs. In the context of sleep study, there are several methods of EEG augmentations [16]. In this work, we adopt two augmentation methods (permutation and crop & resize) which were found effective in [11]. Following are the details of these EEG augmentation methods.

Permutation: The permutation augmentation method changes the temporal order of the EEG signal by dividing it into equal-length non-overlapping segments and randomly shuffling their order. This transformation introduces temporal variability while preserving the semantics of the signal. Given a 30-second epoch x of length L of the EEG signal, we divide x into n_{seg} segments $\{x_1, x_2, \dots, x_{n_{\text{seg}}}\}$. The order is shuffled by shuffling the indices to obtain the permuted signal $\tilde{x}$. In our experiments, we set $n_{\text{seg}} = 10$ in order to divide the input epoch into 10 segments before applying random shuffling.

Crop & Resize: The crop & resize augmentation method crops a randomly selected segment of the epoch and resizes it to the original length of the epoch. This operation introduces variability in the temporal resolution while preserving the global structure.

Given a 30-second epoch x of length L of the EEG signal, a segment of size s is randomly cropped, where $s = \alpha \cdot L$ and α is a scaling factor. The cropped segment x_{crop} is then resized back to length L using interpolation:

$$\text{resize}(x_{\text{crop}}, L). \quad (4)$$

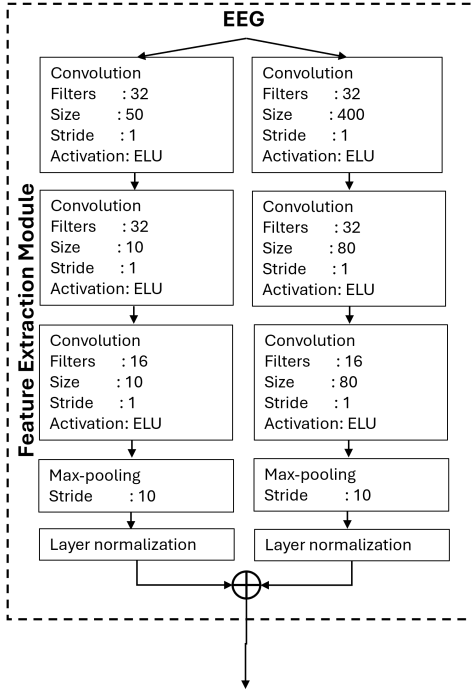


Fig. 2: Architecture of the feature extraction module.

In our experiment, we set the scaling factor $\alpha = 0.2$, meaning that 20% of the signal epoch is cropped and resized to its original length. By applying these augmentations, the model learns to focus on invariant and robust features that generalize well across varying data distributions. The combination of signal permutation and crop & resize ensures sufficient variability in the training data, enabling the contrastive learning framework to capture meaningful representations of EEG signals.

C. Model Architecture

This subsection explains the model architecture used in this research.

1) *Feature Extractor*: We adopt the feature extractor introduced in — having narrow and wide convolutional filters to extract local and global features of the EEG signal, respectively.

The input to the feature extractor is an EEG signal represented as $x \in \mathbb{R}^{n \times C \times L}$, where n is the number of epochs, C is the number of channels, and L is the temporal length of the signal. Figure 2 illustrates the architecture diagram of the feature extractor module showing its specifications. The model accepts a 3-channel EEG signal as input, where each channel is input to a separate feature extractor. The narrow and wide filters capture the patterns in the single channels independently, and the output from three feature extractors are combined using a concatenation layer to form the final representation.

This feature extractor architecture is used for contrastive learning and end-to-end supervised learning addi-

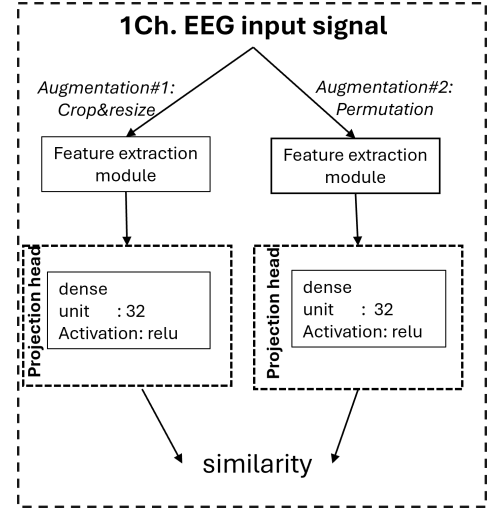


Fig. 3: Contrastive learning model architecture. The feature extraction module’s architecture is given in Figure 2.

tional layers. The details are given in the following subsections.

2) *Contrastive Learning*: As shown in Figure 3, the contrastive learning model architecture has additional layers, called the *projection head*. The input to the contrastive learning model is an augmented pair of the original signal. Generally, the projection head $g(\cdot)$ maps the output of the feature extractor $f(\cdot)$ to z -dimensional space. As shown in Figure 3, the projection head maps the output of the feature extractor module to a 32-dimensional space. During contrastive learning, the feature extractor and projection head are trained together to map positive pairs closer to each other and map positives apart from negatives. The similarity between the projected pairs is used for calculating the contrastive loss (as shown in Equation 2). Upon completion of the training of feature extractor using contrastive learning, the extracted features can be fine-tuned for the downstream task i.e. classification of 5-sleep stages. Algorithm 1 shows the overall contrastive learning algorithm.

3) *Fine-tuning and End-to-end Learning*: Typically, a sleep stage classification model comprises a feature extraction module and a classification module. The model can be trained using labeled data where the initial model weights are randomly initialized. The model weights are learned using the loss function and the gradient descent algorithm. We refer to this scheme of model training in supervised learning settings as *end-to-end* learning as the entire model is trained from the scratch.

On the other hand, *fine-tuning* means training the model for the sleep stage classification using labeled data where the feature extraction module is pre-trained using contrastive learning. Intuitive, the feature extraction module has learned the prominent features from the unlabeled data during contrastive learning, and therefore fine-tuning is expected to achieve better classification and prediction accuracy using a limited number

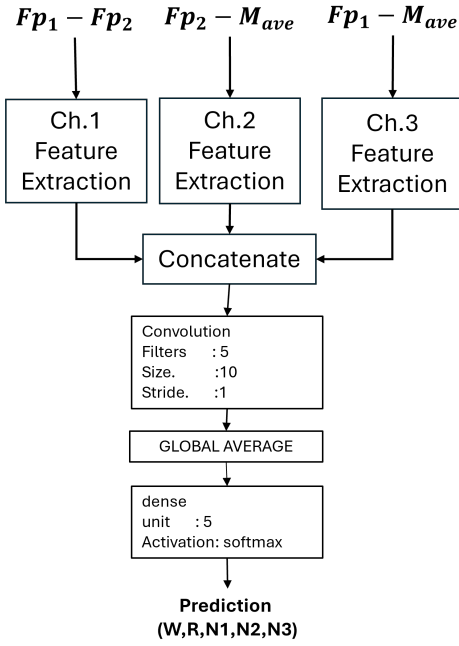


Fig. 4: Sleep stage classification model architecture used for fine-tuning and end-to-end learning.

of labeled data than end-to-end learning.

Both fine-tuning and end-to-end learning use the same architecture, shown in Figure 4, and training pipeline. This enables a comparison of the effectiveness of initializing feature extractors with pre-trained weights learned using contrastive learning with random initialization. By comparing these two approaches, we evaluate whether contrastive learning successfully captures useful representations that improve downstream classification performance.

D. Dataset

This section describes the dataset used in this study, including its basic information, preprocessing steps, and data splitting strategy.

1) *Basic Information*: The sleep EEG dataset used in this study was provided by — — — — — and collected using the in-home EEG device for monitoring sleep. It contains recordings from a total of 398 subjects, comprising 1523 individual records. The EEG signals were recorded at a sampling frequency of 200 Hz, with each record segmented into 30-second epochs. Human sleep experts annotated these epochs into 5 sleep stages - Wake, REM, N1, N2, and N3 - stages according to AASM [1] guidelines.

The dataset includes signals from four electrode positions: $Fp1$ (left frontal), $Fp2$ (right frontal), $A1$ (left mastoid), and $A2$ (right mastoid). These electrodes capture different aspects of brain activity and are commonly used in sleep studies. For this study, three montage were derived from the four raw EEG signals generated by the electrodes of the in-home EEG sleep monitoring device. Hence, the three montages are considered

as the input signal to the model. The montages are derived as follows where M_{ave} is calculated as $(A1 - A2)/2$:

- $Fp1 - M_{ave}$: The potential difference between $Fp1$ and M_{ave} the averaged potential of $A1$ and $A2$.
- $Fp2 - M_{ave}$: The potential difference between $Fp2$ and the averaged potential of $A1$ and $A2$.
- $Fp1 - Fp2$: The potential difference between $Fp1$ and $Fp2$.

2) *Data Pre-processing*: The pre-processing pipeline for EEG signals in this study consists of applying two main steps: bandpass filter and normalization. These steps aim to improve the quality of the signals by reducing noise and ensuring consistency across samples.

Bandpass Filtering: In order to remove noise and artifacts outside the relevant frequency bands, a bandpass filter was applied to the EEG signals. The filter was configured with a sampling frequency of 200 Hz, a passband frequency range of 0.3 Hz to 35 Hz, a stopband frequency range of 0.15 Hz to 70 Hz, a maximum passband loss of 3 dB, and a minimum stopband attenuation of 40 dB. Additionally, a bandstop filter was employed to remove powerline noise at 50 Hz, with stopband frequencies set to 49.9 Hz to 50.1 Hz. This filtering process was applied independently to each of the three EEG channels ($Fp1 - M_{ave}$, $Fp2 - M_{ave}$, $Fp1 - Fp2$).

Normalization: To account for variability in signal amplitude across different subjects and channels, the EEG signals were normalized on a per-channel basis. This normalization step standardized each channel such that all signals were scaled to have zero mean and unit variance as shown in the following the formula:

$$\text{Normalized Signal} = \frac{\text{Signal} - \mu}{\sigma}. \quad (5)$$

The mean and standard deviation for each channel ($Fp1 - M_{ave}$, $Fp2 - M_{ave}$, $Fp1 - Fp2$) were precomputed from the training dataset.

3) *Data Splitting*: The dataset used in this study is divided based on the requirements of contrastive learning and fine-tuning/end-to-end learning. Specifically, the 1361 records are used for contrastive learning, while 162 records were used for supervised learning (fine-tuning and end-to-end learning) with no overlapping records. This ensures that none of the labeled records used for supervised learning were seen during contrastive learning.

For contrastive learning, the 1361 records are split into 90% for the training set and 10% for the validation set. After applying the pre-processing steps, data from the three EEG channels ($Fp1 - M_{ave}$, $Fp2 - M_{ave}$, $Fp1 - Fp2$) are used independently to train single-channel contrastive learning models, resulting in pre-trained feature extractors for each channel.

For fine-tuning and end-to-end learning, the 162 records are split using 5-fold cross-validation strategy.

IV. EXPERIMENTS AND RESULTS

A. Model Training

Table I shows the training parameters for contrastive and supervised (fine-tuning and end-to-end) learning. It may be noted that larger batch size can be beneficial for contrastive learning because there are sufficient negative and positive examples in each batch to learn the data features. Furthermore, all training are performed for 100 training epochs, and in the case of supervised learning, we choose the best model for testing the accuracy.

Parameters	Values
Temperature (τ)	0.5
Batch size (contrastive learning)	512
Batch size (supervised learning)	32
Learning rate (contrastive learning)	1×10^{-4}
Learning rate (supervised learning)	1×10^{-5}
Number of training epochs	100

TABLE I: Training parameters

For the purpose of showing the effectiveness of fine-tuning and end-to-end learning using limited number of labels, the training samples are incrementally increased as 5%, 10%, 20%, 30%, 40%, 50%, 100%. The same training records are used for fine-tuning and end-to-end learning to ensure fair comparison.

1) *Evaluation Metrics*: In order to evaluate our methods, we used accuracy, Cohen's Kappa statistics, macro F1-score and confusion matrix. Furthermore, the reliability of the model's prediction is shown using confidence values predicted by the model. To clarify, the confidence value refers to the probabilities assigned to all 5 stages by the model where the label with the highest probability is considered as the predicted label. The details are given as follows:

- **Confidence Accuracy**: Confidence Accuracy evaluates the average confidence assigned to the true class labels for all test epochs. It is defined as:

$$\text{Confidence Accuracy} = \frac{\sum_{i=1}^N \text{Conf}_{\text{true}}(i)}{\sum_{i=1}^N \sum_{j=1}^C \text{Conf}_{i,j}}, \quad (6)$$

where:

- N is the total number of test epochs,
- C is the total number of classes,
- $\text{Conf}_{\text{true}}(i)$ is the confidence value of the true class for the i -th epoch,
- $\sum_{j=1}^C \text{Conf}_{i,j}$ represents the sum of the confidence values for the i -th epoch.

Since the sum of confidence values for each epoch equals 1, the denominator simplifies to N , the total number of epochs.

- **Confidence Difference**: We compare confidence values of the fine-tuning and end-to-end learning models for each test epoch by taking the difference of the predicted confidence values of the true labels. The term *Positive* means the number of epochs where fine-tuning model assigns a higher confidence to the true label. Similarly,

Negative means the number of epochs where end-to-end model assigns a higher confidence to the true label.

- **U-test for Confidence Significance**: The *Mann-Whitney U-test* is employed to assess the statistical significance of confidence differences between the fine-tuning and end-to-end models. A p -value < 0.05 indicates a statistically significant difference, confirming that one model consistently assigns higher confidence values to the true class labels.

B. Experimental Results

This section presents the experimental results, including the overall comparison of models, confusion matrices, and confidence analysis.

1) *Overall Comparison*: We demonstrate that our contrastive learning based fine-tuning model is label-efficient compared to end-to-end supervised learning. Therefore, we incrementally increased the labeled training data as 5%, 10%, 20%, 30%, 40%, 50%, and 100%, and show the test accuracy averaged over 5-folds of the cross-validation sets. The overall results in terms of accuracy, Kappa score and F1 score are given in Figure 5.

As shown in Figure 5a, the fine-tuning model consistently outperformed the end-to-end model for all proportions of labeled training data. Specifically, when the labeled training data is limited to 20% and 30%, the fine-tuning model achieves an accuracy of 80.85% and 81.62, compared to 79.98% and 80.65% for the end-to-end model, respectively. As the training data proportion increases, the accuracy of both models improves, but the fine-tuning model consistently performs better than end-to-end model, achieving a maximum accuracy of 82.81% at 100% training data, compared to 82.33% for the end-to-end model. A similar trend can be observed in the F1 and Kappa score as shown in Figures 5b and 5c, respectively.

In the overall evaluation, the fine-tuning model showed advantage over the end-to-end model when the labeled training data are limited, with the largest performance improvement observed at 20%, and 30% labeled training data. The results suggest that the pre-training of the feature extractor using contrastive learning enabled the model to learn the prominent features from the unlabeled EEG data from in-home EEG device which allowed the label-efficient fine-tuning of the model for 5-stage sleep classification compared to end-to-end supervised learning.

2) *Analysis of Using Limited Percentage of Labeled Training Data*: We further explore the performance of the two models when 10%, 20% and 30% labeled data are used for training using confusion matrix. As the overall performance comparison showed that 20% and 30% labeled training data have the best performance, we also included 10% labeled data case for more comprehensive comparison.

The confusion matrices are shown in Figure 6. Specifically, rows represent the actual sleep stages (Wake, REM, N1, N2, N3). Columns represent the predicted sleep stages. The values indicate the proportion of samples predicted as each class.

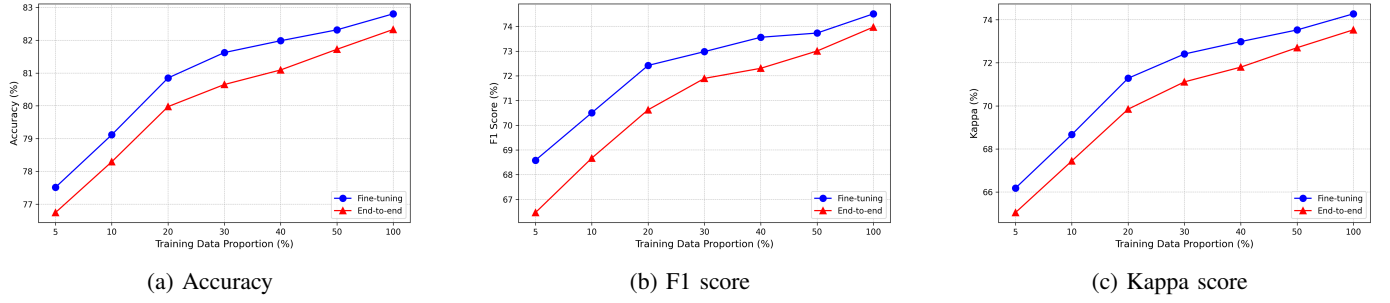


Fig. 5: Average performance comparison between fine-tuning and end-to-end training over 5 folds across metrics (accuracy, F1 score, and Kappa score) with different training data proportions.

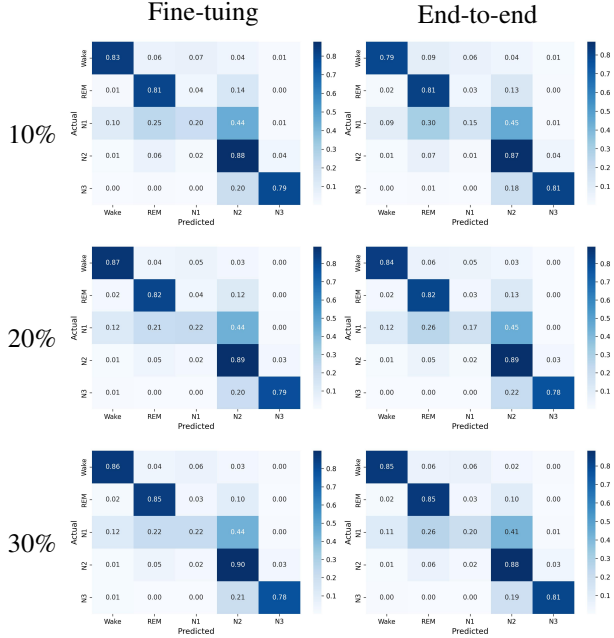


Fig. 6: Confusion matrix visualization of fine-tuning and end-to-end training for different labeled training data proportions (10%, 20%, 30%).

The results from the confusion matrices lead to the following conclusions: For the Wake, REM, N2, and N3 stages, the fine-tuning model maintains higher agreement between actual and predicted labels, particularly in the N2 and N3 stages. This phenomenon occurs due to larger sample sizes of N2 and N3. Under the 10%-30% training data proportions, the fine-tuning model exhibits a lower misclassification rate compared to the end-to-end model, achieving clear performance advantages across most sleep stages.

The visualized confusion matrices further validate the findings presented in the previous section. The fine-tuning model outperforms the end-to-end model across various training data proportions, particularly when labeled data is scarce. By reducing misclassifications and improving prediction accuracy, the proposed contrastive learning method proves to be highly effective in learning discriminative sleep stage features, mak-

ing it particularly suitable for scenarios with limited annotated data.

3) *Confidence Analysis*: We provide further analysis based on epoch-wise evaluation, in contrast with the subject-wise evaluation shown in the previous subsections.

In the analysis, we compared the confidence vectors of each epoch predicted by the two models to examine the confidence accuracy of both models. Based on these confidence vectors, we computed the confidence accuracy of both models, the confidence difference for the true labels of individual epochs, and the statistical significance of these differences through hypothesis testing. The confidence accuracy is calculated using Equation (6), and the comparison between the two models is shown in Figure 7.

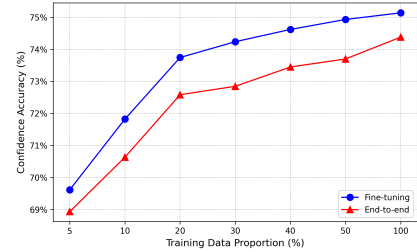


Fig. 7: Confidence accuracy of fine-tuning and end-to-end training across different labeled training data proportions (Aggregated from confidence scores over all epochs in 5-fold cross-validation test sets).

As illustrated in Figure 7, the fine-tuning model consistently achieves higher confidence accuracy compared to the end-to-end model across all training data proportions. Notably, at a training data proportion of 20% and 30%, the fine-tuning model achieves a confidence accuracy close to 74%, outperforming the end-to-end model, which achieves accuracy close to 72.5% and 73%, respectively. This highlights the significant advantage of the fine-tuning model in scenarios with limited labeled data. At 100% training data, the performance gap between the two models narrows. However, the fine-tuning model still maintains a slight advantage, achieving a confidence accuracy of 75.1% compared to 74.2% for the end-to-end model.

These results demonstrate that the fine-tuning model not only assigns higher confidence to the true class labels but also maintains this advantage consistently across different percentages of labeled training data. The performance of the fine-tuning model in lower proportions of labeled training data further confirms the effectiveness of pre-training using contrastive learning to capture meaningful representations, resulting in more reliable and accurate predictions.

Proportion (%)	Positive	Negative	Significance test (<i>U</i> -test)
5	75020	68397	significant
10	80322	63094	significant
20	81032	62383	significant
30	84349	59066	significant
40	83593	59824	significant
50	83794	59664	significant
100	81689	61723	significant

TABLE II: Statistical significance analysis of confidence differences across different proportions. Positive represents number of epochs where fine-tuning confidence is superior, Negative represents number of epochs where end-to-end confidence is superior, and statistical *U*-Test with p -value < 0.05 indicates statistical significance.

In addition to the overview of confidence accuracy, we further analyzed the confidence difference between the two models for individual epochs. Specifically, for each epoch’s true label, if the fine-tuning model achieved a higher confidence value, we marked it as *Positive*; otherwise, it was marked as *Negative*. The statistical results are presented in Table II, where each row corresponds to a different labeled training data proportion. The superior results are highlighted in bold.

Table II suggests that fine-tuning model demonstrates a significant advantage across all the labeled training data proportions. This indicates that our proposed method can consistently predict with higher confidence than traditional end-to-end learning method.

Furthermore, to demonstrate that the confidence differences between the two models is statistically significant, we performed the Mann-Whitney *U*-test. The p -values for the confidence differences across different training data proportions are summarized in Table II. If the p -value < 0.05 , the result is marked as significant. As shown in Table II, the results for all percentages of labeled training data indicate significant differences.

Overall, the fine-tuning the features learned using contrastive learning for sleep stage classification exhibits a clear advantage across all the labeled training data proportions in terms of predicting with more confidence values than end-to-end model. Also, this analysis further validates that the proposed method effectively learns meaningful features, enabling it to provide higher confidence predictions across a larger range of epochs.

V. CONCLUSION

In conclusion, this work proposed contrastive learning framework for label-efficient sleep stage classification model

suitable for multi-channel EEG signal from in-home EEG device. The work demonstrated that contrastive learning can effectively capture important features and patterns from unlabeled EEG data. Therefore, the feature extraction module can be fine-tuned with a small amount of labeled training data in supervised learning settings to achieve better accuracy than training an end-to-end model from scratch. Overall, we showed experimentally that the fine-tuning model significantly outperformed end-to-end supervised learning, particularly when the amount of labeled training data is limited. In addition, the proposed model consistently assigned higher confidence to the predictions.

In future research, this work can be extended by exploring the following directions:

- 1) Multiple positive samples: Incorporating multiple positive samples during the contrastive learning process to enrich the representation space and improve feature generalizability.
- 2) Enhancements of feature extraction module: Increasing the depth and complexity of the convolutional layers in the feature extractor to capture more hierarchical features and expand the capacity of the model.

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